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EXTERNAL LONGITUDINAL WEARABLE — MECHANICAL ARCHITECTURE, POWER, COMMUNICATIONS, AND STANDALONE USE CASES (NON-LIMITING)

A. Mechanical Architecture and Fit Stability (Non-Limiting)

In various embodiments, the external longitudinal wearable comprises a sealed housing containing electronics, a power source, and one or more sensing regions arranged to maintain repeatable coupling during movement and across prolonged wear periods. The wearable may be configured as a C-shaped device or a substantially closed ring. In certain embodiments, the wearable includes: 1. a compliant inner interface configured to maintain stable contact while reducing localized pressure points; 2. one or more internal structural members configured to resist undesired deformation while allowing controlled coupling; 3. alignment cues or geometric features that encourage repeatable placement; and/or 4. low-profile exterior surfaces configured to reduce snagging and improve comfort during sleep. In certain embodiments, the wearable includes replaceable or modular contact interfaces (e.g., sleeves, liners, or removable coupling layers) configured to improve hygiene and repeatability.

B. Materials, Sealing, and Cleaning (Non-Limiting)

In various embodiments, the wearable body comprises medical-grade silicone or other biocompatible elastomer and may include an internal rigid or semi-rigid frame. In certain embodiments, the wearable is configured as a sealed device with ingress protection features. In certain embodiments, the wearable is configured to be cleaned using non-limiting procedures consistent with the housing material, including wipe-down cleaning and/or rinse-safe cleaning. In various embodiments, sensor regions are sealed to maintain stable calibration and resist contamination.

C. Power and Charging (Non-Limiting)

In various embodiments, the wearable includes a battery and power management circuitry configured for all-day and/or overnight operation. Non-limiting power approaches include: 1. rechargeable battery with duty-cycled sensing to extend runtime; 2. charging via a docking base; 3. contact-based or inductive charging; and/or 4. battery health monitoring and charge-state estimation. In certain embodiments, the wearable includes a low-power mode and a wake policy based on time schedule, detected wear, and/or event triggers.

D. Communications and Pairing (Non-Limiting)

In various embodiments, the wearable communicates with a computing system using authenticated wireless transfer, including Bluetooth Low Energy (BLE) in certain embodiments. In certain embodiments, the wearable supports: 1. secure pairing and device association; 2. periodic synchronization of buffered windows; 3. streaming modes for limited durations; and 4. firmware update mechanisms with integrity checks. In certain embodiments, the wearable includes device authenticity checks to reduce unauthorized device participation in reporting and correlation outputs.

E. Standalone Use Cases (Non-Limiting)

In standalone embodiments, Node 1 provides privacy-preserving longitudinal outputs for a single user, including indices, trend curves, variability/stability summaries, and confidence indicators, without requiring other nodes. Non-limiting standalone use cases include: 1. establishing a personal longitudinal reference distribution across days or nights; 2. detecting change-over-time and drift relative to a baseline distribution; 3. producing a repeatability score and stability signature across validated windows; 4. reporting confidence-weighted trends that exclude low-quality windows; and 5. providing context tags and interpretive summaries derived from motion stability, contact integrity, thermal context, and signal continuity indices.

F. Optional Integration Hooks (Non-Limiting)

In optional multi-node embodiments, derived outputs from Node 1 may be correlated with other nodes under consent gating and quality thresholds. The computing system may compute alignment scores, synchrony curves, lag signatures, and confidence indicators using time-aligned windows, without requiring sharing of raw signals in certain embodiments.

G. Variations (Non-Limiting)

Any embodiment described herein may omit sensors, add sensors, substitute sensor modalities with equivalents, alter housing geometry, alter gap size for C-shaped embodiments, alter duty cycles, and/or alter processing distribution (on-node, on-device, cloud) while maintaining the functional intent of longitudinal collection, validity gating, and privacy-preserving outputs.